

1. Introduction

The Integrated Library Management System (ILMS) is a software solution designed to automate the manual processes involved in managing a library's daily operations. Traditionally, libraries relied on physical registers and card catalogs to track book inventory, member records, and borrowing history. This project aims to transition these operations into a robust relational database system, ensuring data integrity, security, and efficient retrieval.

2. Objectives

- **Automation:** To automate the process of book issuance, returns, and fine calculations.
- **Data Centralization:** To maintain a centralized repository for books, authors, publishers, and members.
- **Inventory Management:** To provide real-time updates on book availability and stock levels.
- **Search Optimization:** To enable users to search for books by various parameters like title, ISBN, or genre.
- **Security:** To implement role-based access control (Admin & Member).

3. Problem Statement

Manual library management systems suffer from several critical issues:

- High risk of data redundancy and inconsistency across paper records.
- Difficulty in tracking overdue books and calculating fines accurately.
- Inefficient search mechanisms making it time-consuming for members to find resources.
- Lack of data backup and high physical storage requirements for records.

4. Methodology

The development will follow a structured Database Life Cycle (DBLC):

1. **Requirement Analysis:** Identifying the needs of librarians and students.
2. **Conceptual Design:** Creating an Entity-Relationship (ER) Diagram to map data structures.
3. **Logical Design:** Mapping the ER diagram to relational tables and normalizing them to 3NF.
4. **Physical Design:** Implementing the schema in a Database Management System (e.g., MySQL or PostgreSQL).
5. **Testing:** Running SQL queries to ensure data retrieval performance and constraint validation.

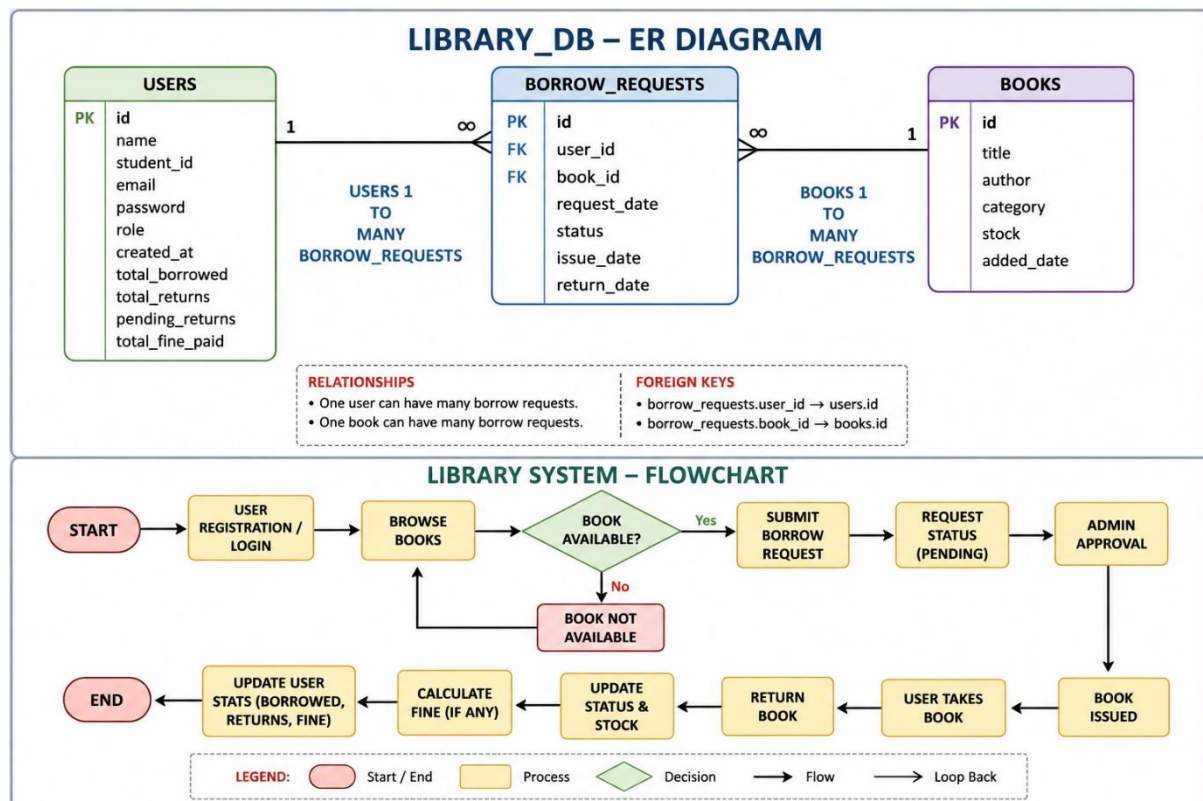
5. Database Design

The system will be built around the following core entities and relationships:

Key Entities:

- **Books:** Title, Language, and Publication Year.
- **Authors:** Library author University Central
- **Members:** Library Member ID, Name, Email, and Membership Type.
- **Transactions:** Tracks Issue Date, Due Date, and Return Date.

The relationship follows the rule: **Member (1) → (N) Transactions ← (1) Book**. Normalization will ensure that partial and transitive dependencies are removed to maintain data integrity.



6. Requirements

Software Requirements

- **Database:** MySQL / PostgreSQL
- **Programming Language:** HTML (Markup) & PHP (Scripting)
- **Operating System:** Windows 10/11 or Linux

Hardware Requirements

- **Processor:** Dual-core 2.0 GHz or higher
- **RAM:** 4 GB Minimum
- **Storage:** 100 MB for database file growth

7. Expected Outcome

Upon successful completion, the ILMS will provide:

- A fully functional relational database schema with enforced referential integrity.
- A user-friendly interface for librarians to manage inventory.
- Automated fine calculation reports for overdue materials.
- A reduction in administrative overhead by approximately 60%.

8. Timeline

Phase	Task Description	Duration
Phase 1	Requirement Analysis & ER Modeling	Week 1
Phase 2	Schema Normalization & SQL Implementation	Week 2
Phase 3	Backend Logic & Query Optimization	Week 3
Phase 4	Frontend Integration & Final Testing	Week 4

9. Conclusion

The proposed Library Management System is a vital step toward digitizing educational resources. By leveraging a well-designed database, the system will not only solve existing manual errors but also provide a scalable platform for future enhancements, such as digital e-book integration and automated notification systems.